

Fat Cell Structures & Generalized Algebraic Theories

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Generalized Algebraic Theories (GATs)

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Sort declarations, operation (function) declarations, equation declarations

Generalized Algebraic Theories (GATs)

Graded semigroups (or graded rings):

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$$m, n : M \vdash m + n : M$$
$$m, n, o : M \vdash (m + n) + o = m + (n + o) : M$$

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- Later declarations use complex expressions *generated* by earlier ones

Generalized Algebraic Theories (GATs)

Cell complexes:

Generalized Algebraic Theories (GATs)

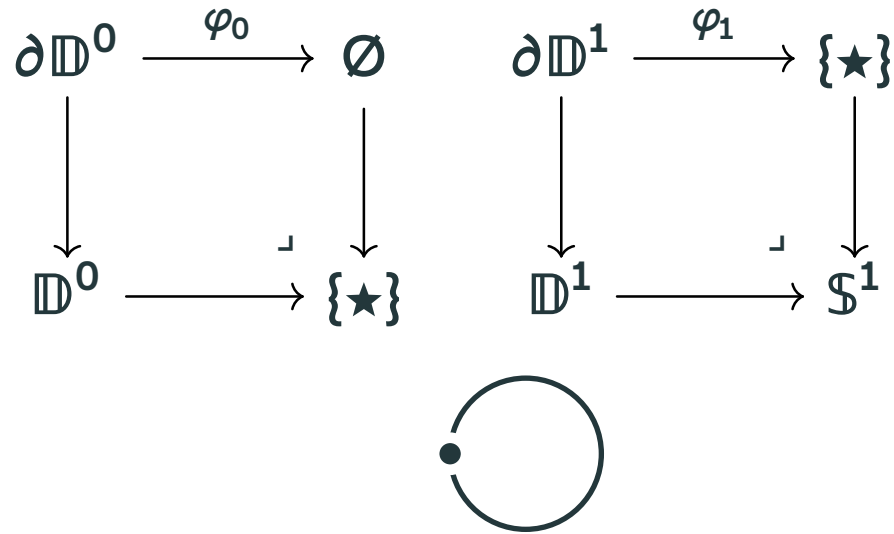
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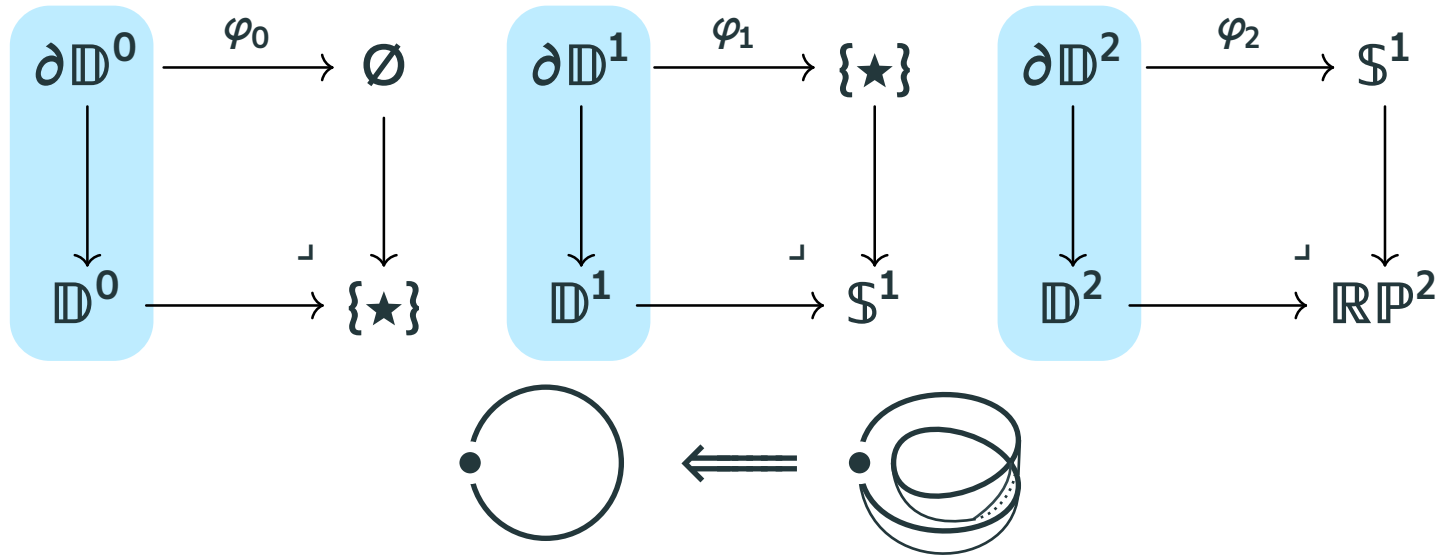
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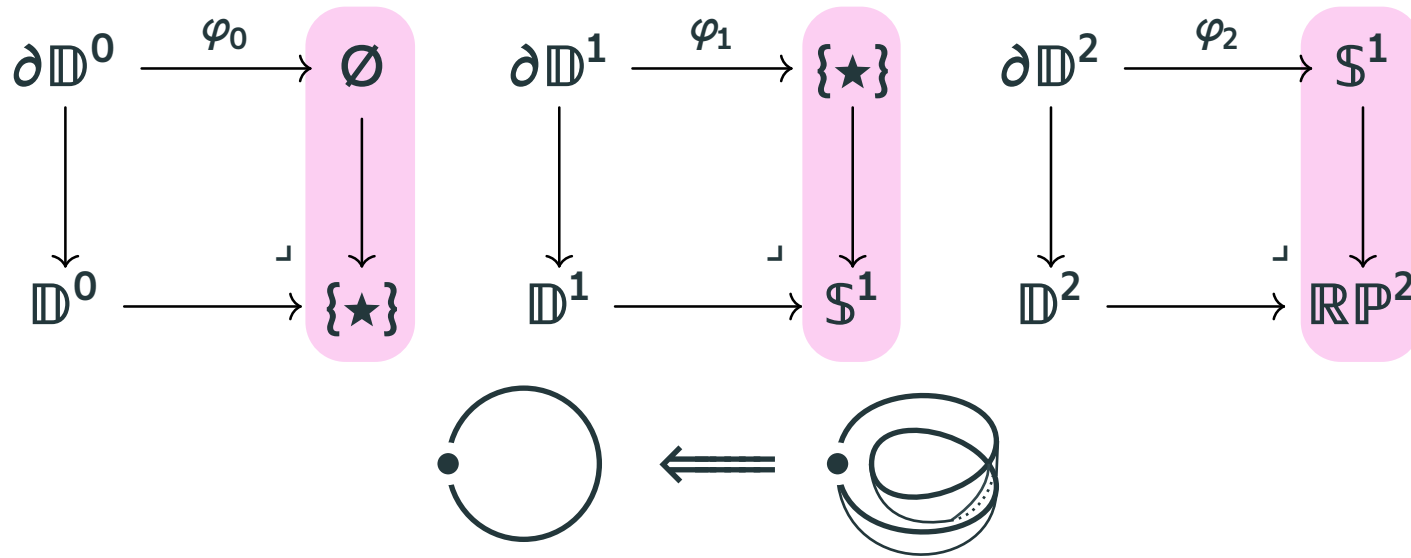
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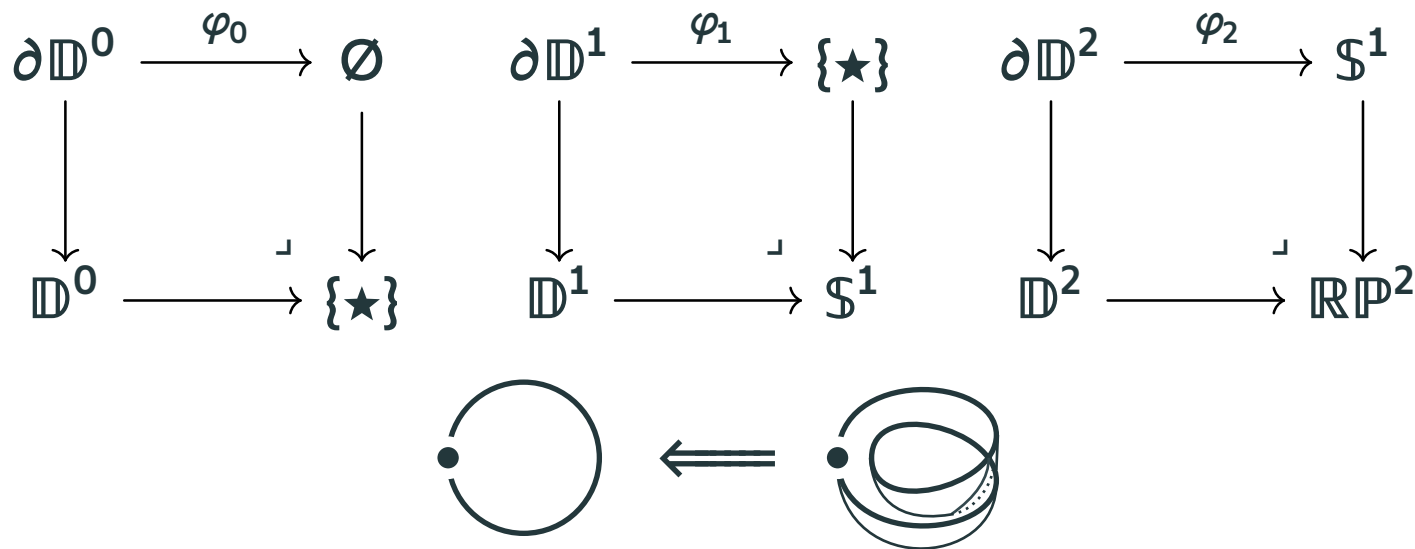
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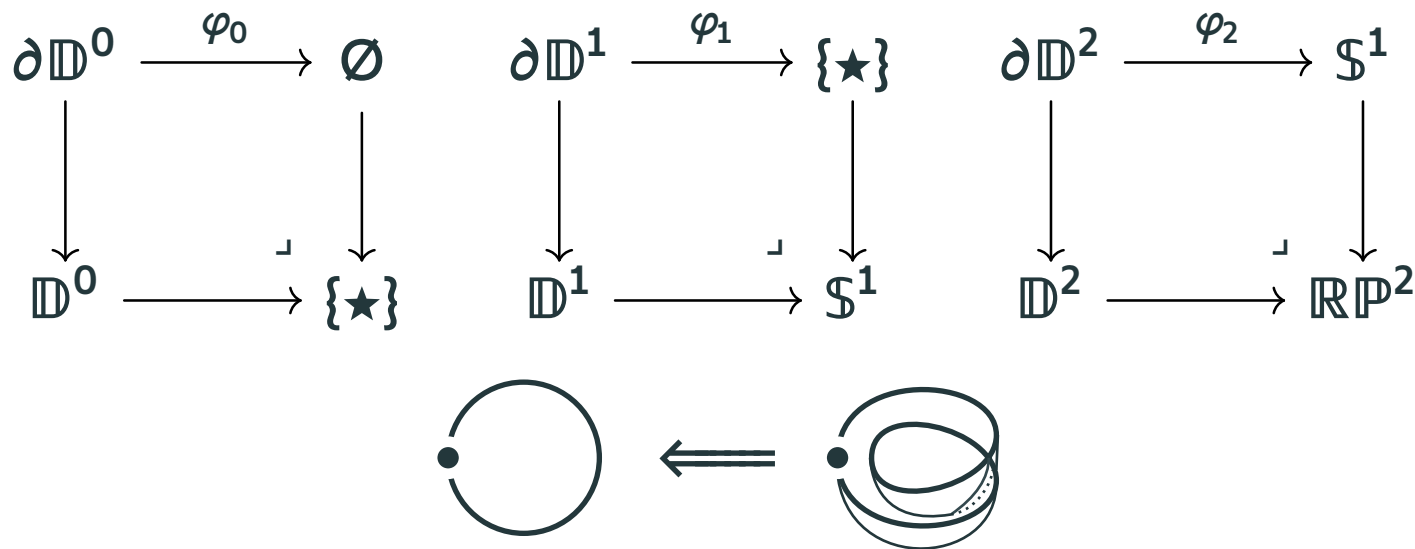
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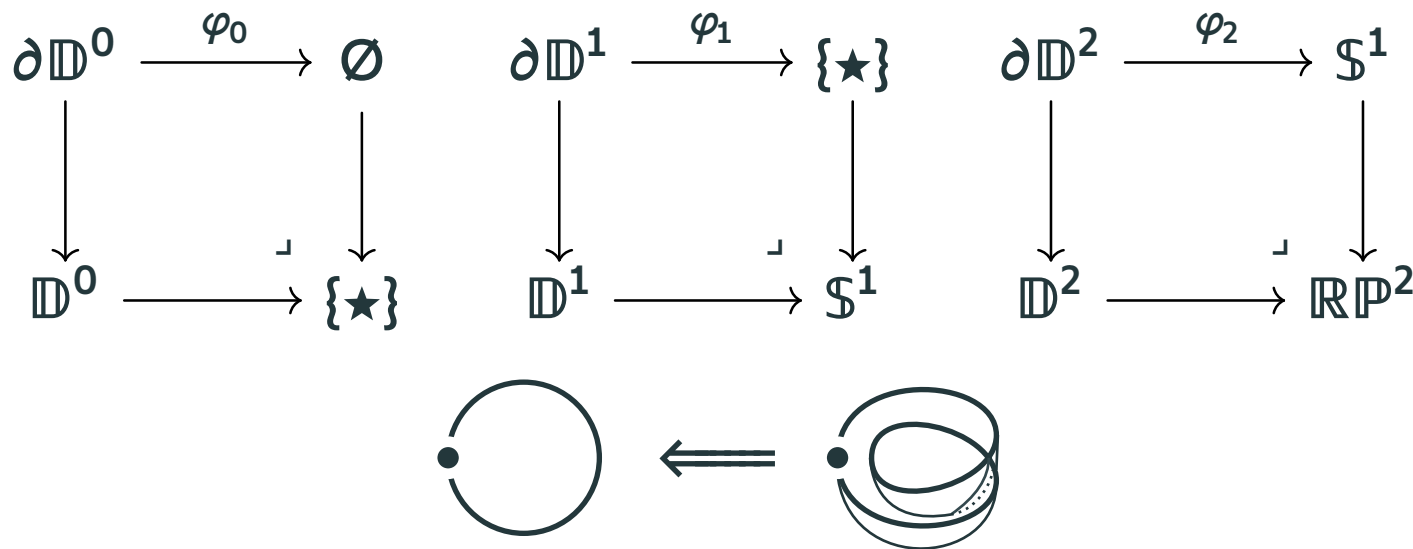
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How do we formalize this intuition? How do we rigorously define GATs at all?

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- "A *generalized algebraic theory* is a well-formed pretheory."

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Consider another type theory ("Theory of Signatures") where

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and we have* Π and extensional equality types.

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We can define a (finite) GAT as a context in ToS.

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Define *signatures* Σ (inductively) together with the category **CwF** $_{\Sigma}$ of cwfs equipped with a Σ -structure (recursively).

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$\mathbf{CwF}_{\Sigma, \Gamma}$ defined in terms of $\mathbf{CwF}_\Sigma$ and Γ .

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- Let **C** be the free cwf generated by an object, i.e. $\text{hom}(\mathbf{C}, \mathcal{A}) \cong \text{Ob}(\mathcal{A})$.
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We can define a finite GAT as a **finite cell complex** over the morphisms

$$\{\mathbf{C} \rightarrow \mathbf{S}, \mathbf{S} \rightarrow \mathbf{T}, \mathbf{T}^{(2)} \rightarrow \mathbf{T}\}$$

The BCDE construction, now with arrowsTM

$$\begin{array}{ccc} \mathbf{c}/\mathbf{CwF} & \xleftarrow{\mathcal{C} \mapsto \Gamma_{\mathcal{C}}} & \mathbb{C}(\Sigma)/\mathbf{CwF} \\ \uparrow & & \uparrow \\ \mathbf{s}/\mathbf{CwF} & \xleftarrow{\quad} & \mathbb{C}(\Sigma, \Gamma)/\mathbf{CwF} \end{array} \qquad \begin{array}{ccc} \mathbf{C} & \xrightarrow{[\Gamma]} & \mathbb{C}(\Sigma) \\ \downarrow & & \downarrow \\ \mathbf{S} & \xrightarrow{\quad} & \mathbb{C}(\Sigma, \Gamma) \end{array}$$

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Theorem. $\mathbf{CwF}^{\text{op}}$ is a cwf! It is a model of the theory of signatures.

Theorem. $\mathbf{CwF}_{\text{fin}}^{\text{op}}$ is equivalent to the syntactic model of ToS.

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Small object argument: transfinite compositions* of pushouts

Fat small object argument: a well-founded *poset* of cell attachment.

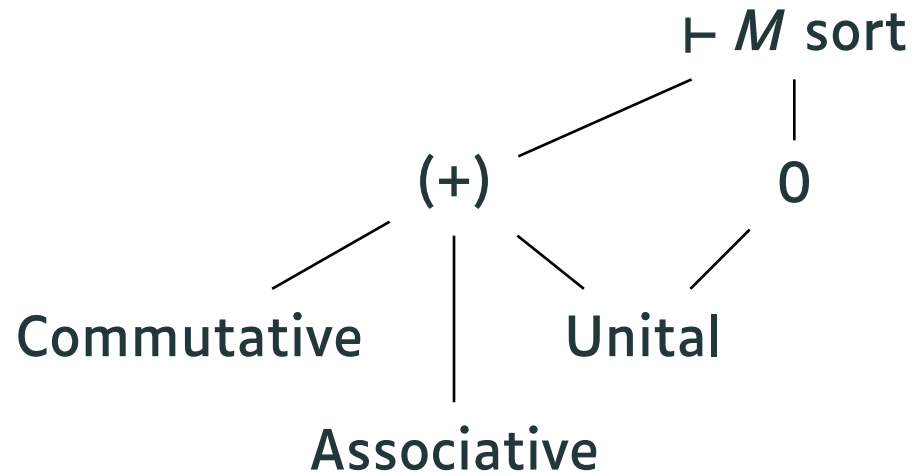
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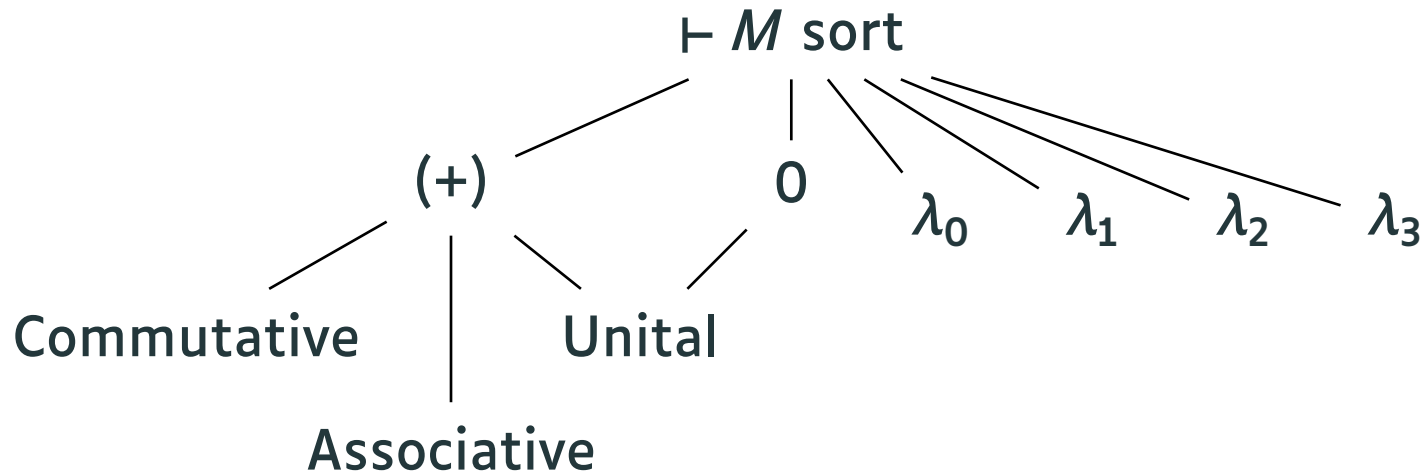
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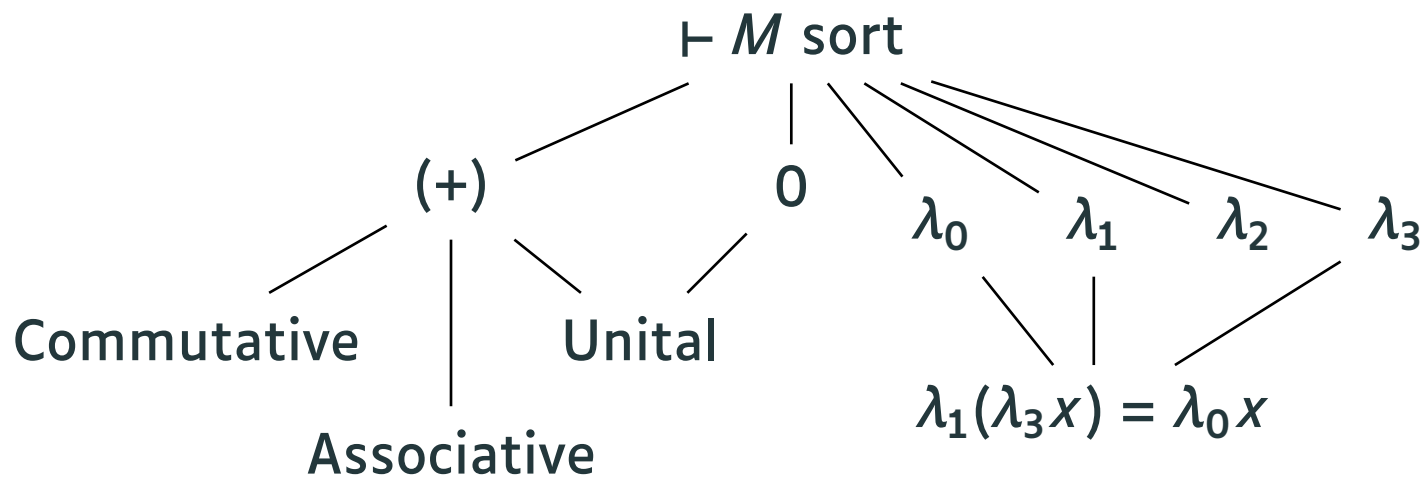
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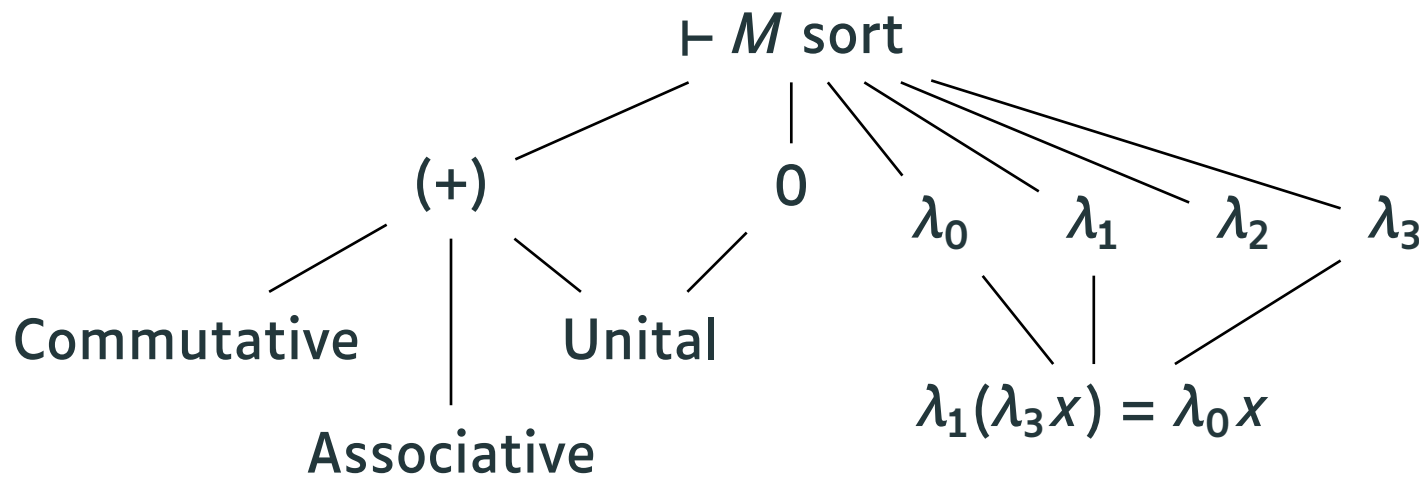


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Principle: Every declaration makes sense given everything above it.

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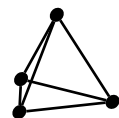
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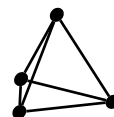
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“Width” vs “Height” of posets

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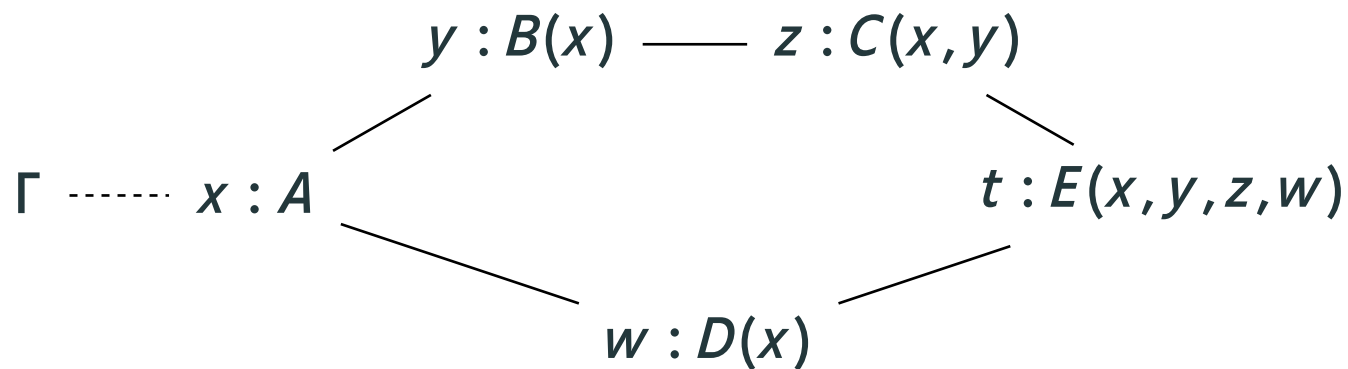
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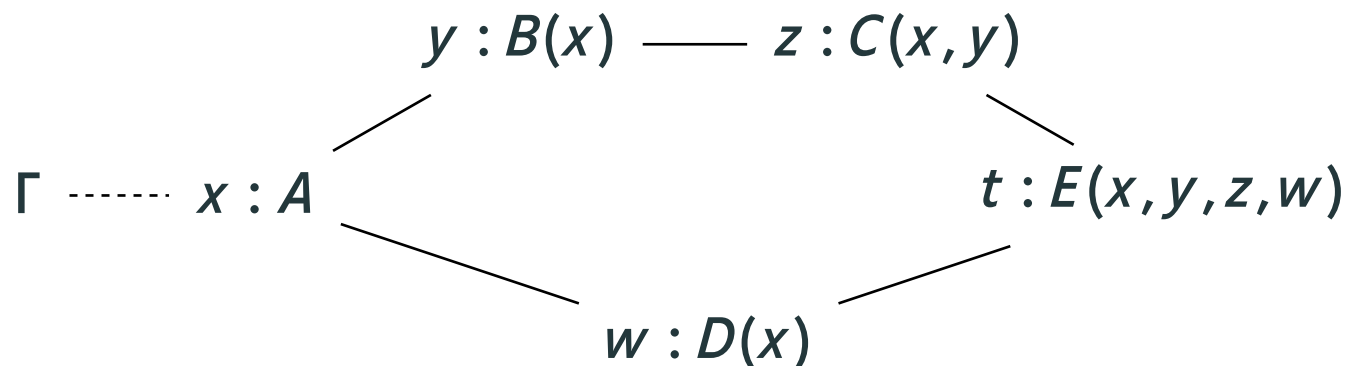
Infinite *depth*: We can similarly go transfinite, see Martínez and Henry [MH25].

Infinite *width*: Similar to Kovács and Kaposi's work on large QIITs [KK20].

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Poset P gives a presheaf map $\mathbf{Tms}_P \rightarrow \mathbf{Tel}_P$, require to be representable.

If $\mathbb{P}$ is a collection of well-founded posets, define $\mathbb{P}$ -CwF to be a cwf supporting P -shaped context extensions for all $P \in \mathbb{P}$.

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 $Z : \text{ob}$ $\searrow$ $g : \text{hom}(Y, Z)$ $\vdash g \circ f : \text{hom}(X, Z)$

$\vdots$

Our contribution

- Explicitly wrote down the cell complex view of GATs
- Fat/infinite presentations of (ordinary) GATs
- Fat/infinity GATs and fat categories with families
- A version of the Gabriel–Ulmer duality following Frey [Fre25]

Further questions

- Compare *finitely presented* GATs to GATs that are *finitely presentable* (compact) when considered as objects in **CwF**
- Generalize well-founded posets to directed categories
- Good examples of infinitary, truly dependent algebraic theories
- GATs are potentially useful for specifying homotopy-coherent structures with only set-level constructions

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